

# An Integrated Assistive System for Visually Impaired Individuals Using Autonomous Navigation, Deep Learning Currency Recognition, and Real-Time GPS Track

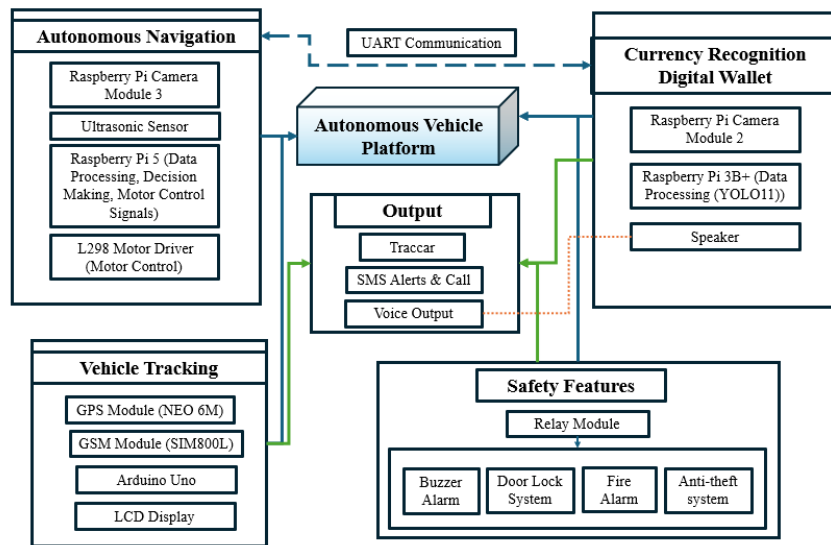


Fig. 1. Overall system architecture.

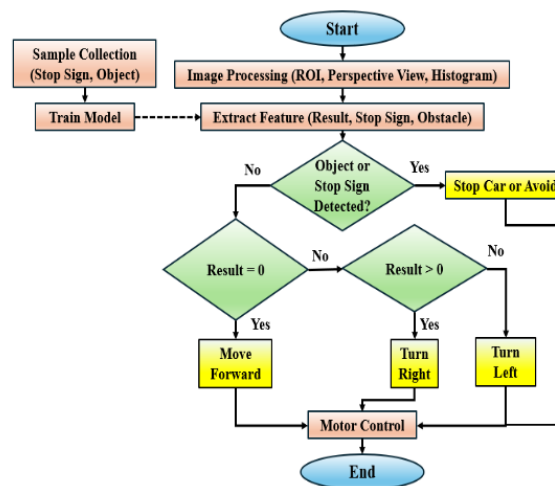


Fig. 2. Operational flow diagram of the autonomous car.

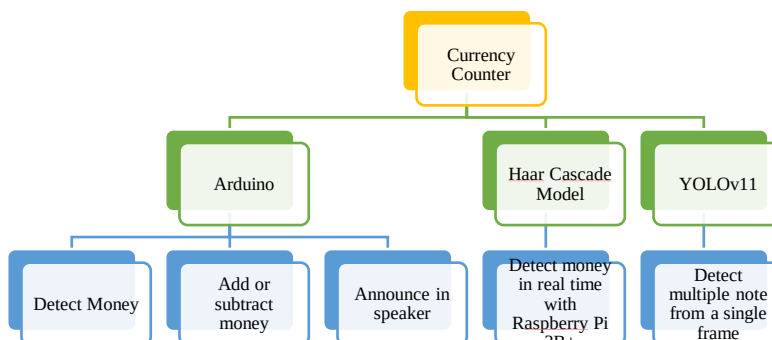


Fig. 3. Architecture of the currency counter subsystem.

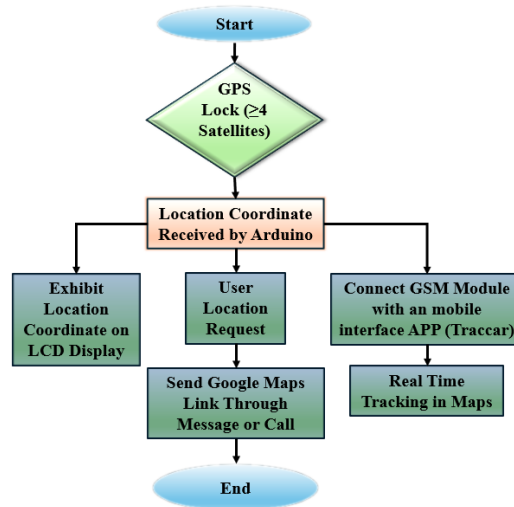


Fig. 4. Operational flow diagram of GPS tracking subsystem.



Fig. 5. Autonomous prototype in customize track.



Fig. 6. Image Processing with Raspberry Pi 5.



Fig. 7. Stop sign and obstacle detection with YOLOv8

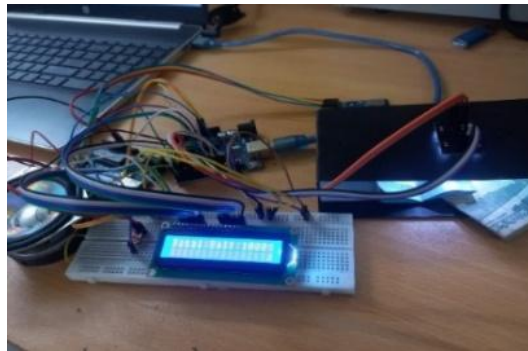


Fig. 8. Money detection using a color sensor and Arduino Uno.



Fig. 9. Money addition and subtraction operation in a currency counter with Arduino.

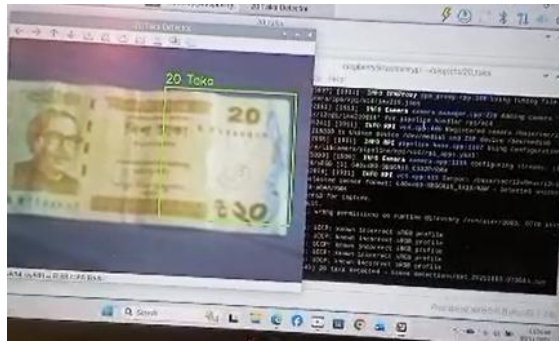


Fig. 10. Money Detection with Haar Cascade Model.



Fig. 11. Money detection with the YOLOv11 model.



Fig. 12. Satellite locking, location with LCD display and Arduino Uno.

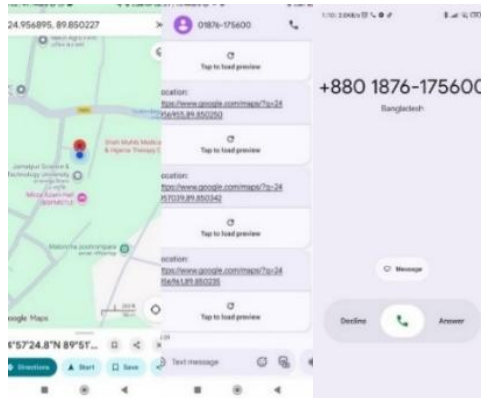


Fig. 13. Sending SMS and calling the user's family member with a GSM module.

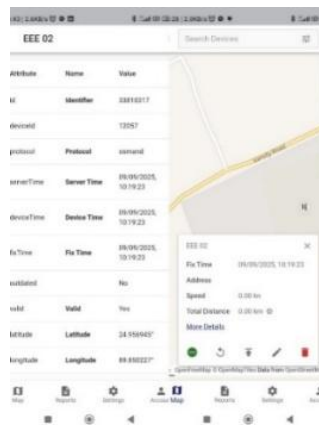


Fig. 14. Traccar app interface with real-time location.



Fig.15. Additional features along with the relay module.